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Surface Modified Boron Nitride as A Filler to Achieve High Thermal Stability of Polymer Solid-State Lithium Metal Batteries

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ABSTRACT

Solid polymer electrolytes with low density, low cost and excellent processability have been the ideal choice for researchers. However, their thermal stability and mechanical properties are so inadequate that it is difficult for them to withstand a sufficiently high temperature and to effectively suppress Li dendrites. PEO cannot inhibit dendrite growth by itself, especially when the battery is operated at a higher temperature (>80°C), the mechanical properties of PEO are significantly reduced. At high temperatures, the growth of lithium dendrites is an important factor affecting the cycle life of Li metal batteries. Therefore, polymer electrolytes with high mechanical strength and excellent thermal stability can overcome the problems associated with Li metal anodes and realize the application of rechargeable Li batteries in high-temperature situations. Here, we prepared a polyethylene oxide (PEO) composite polymer electrolyte (CPE) with thermal stability, high mechanical strength with the modification of hexagonal boron nitride nanosheets (h-BNNS). The surface modification of h-BN enhanced the interface interaction between h-BNs and improved the thermal conductivity, thereby greatly enhancing the thermal stability of the battery. At the same time, the surface modification of h-BN enlarges the effective specific surface area of the filler, which can promote the interaction with the Lewis acid-base of anion, helping the dissociation of Li salts to release more Li^+ ion. This type of "One Sagittal Two Piercing" effect enabled the Li/$LiFePO_4$ full battery with PEO/LiTFSI/SiO_2@BNNS (PLSB) composite electrolyte to perform long-term cycles (900 cycles). Even when the battery was operated at a temperature as high as 150°C, it could cycle stably for more than 1000 cycles at a rate of 2 C, making the lithium metal battery suitable for operation at higher temperatures.

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INTRODUCTION

Lithium metal batteries have the advantages of high energy density, high theoretical capacity (3860 mAh g^{-1}) and low potential (−3.04 V compared to standard hydrogen electrodes), so they are widely regarded as the most potential next-generation rechargeable energy storage equipment with the greatest potential.[1-3] However, during the battery cycle, the lithium metal anode will react with the organic liquid electrolyte and cause uneven lithium-ion deposition, low coulombic efficiency and unsatisfactory electrochemical performance. In the worst case, it may cause combustion or explosion.[4-9] In addition, the flammable nature of the organic liquid electrolyte makes decomposition easy under high temperature or pressure, which severely limits the use environment of the battery.[10-13] The above-mentioned problems are the main problems to be solved in the commercialization of lithium metal batteries. Fortunately, because of their non-flammability and high safety, solid-state electrolytes (SSEs) including oxides, sulfides, halides, and polymers are considered safe alternatives to liquid electrolytes. [14] Compared to liquid electrolytes, their good mechanical properties can also effectively alleviate the growth of lithium dendrites. [15-19]

Solid polymer electrolytes (SPEs), due to their excellent physical properties such as flexibility, film-formation and processability, conform to a certain extent to the development trend for light weight, safety.[20] As the earliest and most widely studied polymer electrolyte,[21] PEO has the advantages of low cost, light weight, and high solubility to lithium salts. However, poor mechanical properties as well as the low ionic conductivity of PEO-based SPEs are harm to alleviate the growth of lithium dendrites. In addition, although it has been shown anecdotally that a PEO SPE can work at temperatures as high as 120°C,[22] the charging and discharging behavior of the battery is not steady, and the fluidity of PEO increases and the mechanical strength deteriorates, which can easily lead to short-circuit of the battery. These shortcomings limit their utility in solid-state batteries. In the past several years, much work has been done to improve the physical and chemical properties of PEO-based SPEs.[23, 24] Many methods have also been studied, including the formation of copolymers, the formation of cross-linked polymers, and the addition of complex salts, plasticizers, and inorganic fillers. It

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is worth mentioning that blending inert inorganic fillers (TiO_2,[25, 26] Al_2O_3,[27] Fe_2O_3,[28] $ZnAl_2O_4$,[29] CeO_2,[30] MMT,[31] and $Mg_2B_2O_5$[32]) into PEO-based SPE is a flexible, simple and effective method. However, the method of adding nanoparticles directly to SPE will cause problems such as agglomeration between particles and poor compatibility between polymers and fillers, which limit the improvement in ionic conductivity.[33] Although the above methods improve the ionic conductivity of the PEO-based solid electrolyte and the ability to inhibit the growth of lithium dendrites, the high temperature resistance problem of the PEO-based solid electrolyte is still unsolved, which cannot meet the practical application of high temperature lithium metal batteries. It is necessary to mention that due to the low thermal conductivity of polymer electrolytes, the thermal gradient of the battery will increase during use.[34-36] As a result, when the internal temperature of the battery reaches the melting point of the polymer, the melting of the polymer electrolyte may cause a potential internal short circuit.[37] Though it has been reported that the incorporation of ionic liquids (ILs) into the polymer matrix allows the battery to operate at high temperatures, the incorporation of IL into the polymer matrix will reduce the mechanical properties of the polymer electrolyte. In addition, high temperatures may cause IL leakage.[38] Thermally conductive polymer separators and electrolytes have been used in lithium-ion batteries to promote heat dissipation and thermal stability.[39] However, due to the addition of large amounts, which reduces the percentage of the electrolyte mixture and aggravates the random distribution of the fillers, the thermal resistance of the electrolyte also increases. Therefore, finding low-cost and effective methods to enhance the thermal stability of SPEs and to improve the dendrite and thermal runaway problems of solid-state lithium metal batteries (SSLMBs) is an urgent problem.

h-BN has an extremely high temperature resistance and chemical stability and is often used as a refractory material. Therefore, the use of boron nitride nanomaterials and polymer composites can greatly improve the thermal conductivity of the polymer and can also maintain the electrical insulation of the polymer. Jong-Chan Lee et al.[40] reported the use of perfluoropolyether (PFPE) functionalized 2D boron nitride nanoflakes (BNNFs) as a multifunctional additives to inhibit lithium dendrites in GPEs

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and improve the electrochemical performance and safety performance of SSLMBs. Although h-BN has the above-mentioned strength, its potential as a CPE additive has not been fully exploited. Its formation of chemical bonds with the polymer matrix is difficult, so the compatibility is very poor. The uneven dispersion of the filler in the polymer matrix will cause a higher heat conduction resistance at the interface, thereby reducing the thermal conductivity, Therefore, its excellent thermal stability is not completely reflected in the composite electrolyte.

In this study, we report a simple and feasible method to synthesize a PLSB CPE. The SiO_2@BNNS composite material forms a thermal pathway in the polymer system through the casting method, which greatly improves the thermal stability of SPE and has a positive effect on the ionic conductivity and mechanical strength. The surface modification of BNNS enhances the interface interaction between BNs, thereby reducing the thermal resistance and can effectively improve the thermal conductivity with a relatively low filler content and extremely improve the thermal stability of the CPE. In the meantime, the h-BN with a large effective surface area can promote the interaction with the Lewis acid base of the anion, which is benefit for the dissociation of lithium salts, thereby improving the ion conductivity. In addition, due to the mechanical strength of SiO_2@BNNS, the formation and growth of lithium dendrites will be inhibited. All solid-state lithium batteries (ASSLBs) using PLSB CPE have proven good performance. A Li/$LiFePO_4$ battery assembled with a PLSB CPEs showed high initial capacity (162.3 mAh g^{-1}) at 0.2 C, the specific discharge capacity at 150°C and 2 C rate exceeded 150 mAh g^{-1} (153.5 mAh g^{-1}), and the discharge capacity retention was 93% after 100 cycles at 150 °C. At the same time, Li/CPE/Li can be stable for 900 h at 150°C and 0.4 mA cm^{-2}. These stable battery performance show that the PLSB CPE membrane has the potential to be suitable for LMBs operating at high temperatures.

EXPERIMENTAL SECTION

Preparation of BNNS:

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The BNNS were obtained by sonication and freeze-drying. First, 1g of h-BN powder was added to 50 ml of *N*-methyl pyrrolidone (NMP) solvent and sonicated for 72 h, and the supernatant separated. Then the NMP solvent was replaced with absolute ethyl alcohol and sonication continued for 24 h. Next, the prepared BNNS solution was freeze-dried to obtain the BNNS products.

Preparation of SiO_2@BNNS composite material:

An appropriate amount of BNNS was added to a mixed solution of 70 mL of absolute ethanol and 30 mL of deionized water, stirred mechanically for 30 min and dispersed ultrasonically for 10 min. It was then placed in a 60 °C constant-temperature water bath and magnetically stirred to obtain a BNNS suspension. An appropriate amount of ammonia was added to the suspension and mixed well. A 9-mL volume of tetraethyl orthosilicate (TEOS) was divided into three equal parts, and one part added slowly to the suspension every 2 h to ensure that the added TEOS is fully reacted. The total reaction time added is 6 h. After the reaction, the product was centrifuged at 3000 rpm for 10 min and washed three times with absolute ethanol to remove reaction residues, such as ammonia. After centrifugation, the product was dried in a vacuum drying oven at 60 °C for 24 h to obtain the SiO_2@BNNS composite powder.

Preparation of composite polymer electrolyte (CPE):

PEO (molecular weight: 300,000; Aladdin), lithium bis(trifluoromethanesulfonic)imide (LiTFSI; Aladdin) and anhydrous acetonitrile (CH_3CN; Aladdin) were used without further purification. A certain amount of LiTFSI, SiO_2@BNNS and PEO were added to CH_3CN and stirred for 12 h to form a homogeneous solution as the CPE precursor. The CPE film was prepared by casting the precursor in a customized mold. Finally, the CPE was dried under vacuum at 60 °C for 24 h to completely remove the acetonitrile solvent. It was then transferred to a glove box filled with argon (O_2 and H_2O were at ppm-levels, usually <0.1 ppm) to remove solvent and moisture. The studied PEO/Li salt molar ratio was 8:1, and the SiO_2@BNNS content in the CPE varied from 0 to 8 wt %.

Material characterization:

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Scanning electron microscopy (SEM) and transmission electron microscopy (TEM)were used to study the morphology. Elemental analysis was conducted on an energy dispersive X-ray (EDX) spectrometry in conjunction with SEM and TEM. The CPE composition was studied by X-ray diffraction (XRD), the sample scanned in the 2θ range of 10−80°. Thermal gravimetric analyzer (TGA) was used to analyze the thermal stability of the polymer electrolyte at a scan rate of 10°C min^{-1} in a temperature range of 20−800°C under N_2 atmosphere. The Fourier transform infrared (FT-IR) spectrum were obtained on a Bruker Vertex 70 FT-IR spectrometer. The mechanical stability of the CPE was measured by an electronic stretching machine.

Electrochemical measurement:

The ionic conductivity of the CPE was evaluated by electrochemical workstation (CHI660E) through electrochemical impedance technology, in which the temperature range was set at 30−80℃ with a frequency range of 10^{-2}-10^{6} Hz. The SSE membrane was sandwiched between two parallel stainless steel disks. The ionic conductivity (σ) of the electrolyte membrane was calculated by equation (1), where R_b represents the resistance of the electrolyte, L represents the thickness of the solid electrolyte (cm), and S represents the area of the solid electrolyte (cm^2):

$$\sigma=\frac{L}{R_b S} \quad (1)$$

The electrochemical window of the CPE was tested by linear sweep voltammetry (LSV) in a cell that sandwiched the electrolyte film between stainless steel and lithium metal. LSV was performed on the CHI660E electrochemical workstation at 60 °C at a scan rate of 1 mV s^{-1}.

The Li-ion transference number (t_{Li^+}) of the SSE was measured in a Li/SSE/Li cell with DC polarization. A voltage of 10 mV (ΔV) was applied and EIS spectra of cell before and after polarization were obtained from 10^{-2}−10^{6} Hz. The (t_{Li^+}) of the electrolyte membrane was calculated by equation (2):

$$t_{Li^+}=\frac{I_s \Delta V - I_0 R_0}{I_0 \Delta V - I_s R_s} \quad (2)$$

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Where I_0 and I_s are the initial current and steady state current, ΔV is the polarization potential at 10 mV. R_0 and R_s are the initial and final interface resistances, respectively.

The electrochemical test was carried out at 60°C with a coin cell consisting of a lithium metal anode, SSE and $LiFePO_4$ cathode (working electrode). The working electrode was composed of active material ($LiFePO_4$) with a weight ratio of 8:1:1, conductive carbon black and PVDF used as a binder. $LiFePO_4$, carbon black powder and PVDF were mixed in a mortar, and N-methyl-2-pyrrolidone (NMP) was used as a solvent. The slurry mixture was coated onto carbon-coated aluminum foil, dried in a blast drying oven at 60°C for 6 h, and then vacuum dried at 100°C for 12 h. The active substance loading was 1.5−2.0 mg cm^{-2}. The battery was assembled in a glove box filled with argon (O_2 and H_2O were at ppm-levels, usually <0.01 ppm). A land coin cell testing system was uses to charge/discharge the Li/SSE/$LiFePO_4$ batteries between 2.7 and 3.8 V.

The symmetric cell was prepared by sandwiching the polymer membrane with two Li foils (1.54 cm^2). The cell was measured in a land coin cell testing system in an oven at 60 ℃ by changing the cycling time and current.

RESULTS AND DISCUSSION

PLSB CPE was prepared by solvent casting method. The preparation process of CPE is shown in Figure 1. In the SEM images of h-BN (Figure S1), it can be seen that h-BN is a well-dispersed ultra-thin nanosheet with diameters in the range of 300−400 nm. In Figure S2, the elemental spectrum of h-BN reflects the uniform distribution of nitrogen and boron in h-BN. The SEM image in Figure 1b shows that SiO_2 nanoparticles were uniformly dispersed on the surface of the BNNS, the inset shows the image of SiO_2@BNNS under the higher magnification. The elemental map in Figure S3 reflects the uniform distribution of silicon and oxygen on the BNNS, which indicates that the obtained samples are of high quality. To further verify the structure and composition of SiO_2@BNNS, XRD was carried out, as shown in Figure 1d, in which it can be seen that the h-BN peak shape is sharp, the structure is complete, and no characteristic peak of SiO_2 is present in the spectrum. Because the main peak of h-

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BN is too intense, only one can be seen near 23° for the low peak package, if the 10−30° part of the spectrum is partially magnified, it can be observed that there is an obvious complex peak between 20−30°. This is the characteristic 'steamed bun' peak of amorphous SiO_2, indicating the formation of the reaction SiO_2 in the reaction in an amorphous form. Figure 1c is the TEM image of SiO_2@BNNS. BN has a flaky structure, and a layer of nanoparticles is affiliated to the surface of the flake-shaped BNNS. According to the EDX spectrum, the four elements of Si, O, N, and B existed simultaneously and were evenly distributed. Combining SEM, XRD and EDX analysis can further confirmed that these nanoparticles were SiO_2 particles, that is, a core-shell structure with flake h-BN as the core and nano-SiO_2 as the shell. The reason for the combination of SiO_2 and h-BN may be that TEOS first undergoes a hydrolysis reaction under alkaline conditions to form silanol, and the hydroxyl groups of silanol will form hydrogen bonds with the hydroxyl groups on the surface of BNNS. Under heating, the hydroxyl group of silanol and the hydroxyl group on the surface of h-BN will undergo a condensation reaction, and finally SiO_2 and h-BN are chemically bonded together, which is not easily destroyed. The well-distributed SiO_2 has a large effective surface area, which may can promote the interaction with the Lewis acid-base of the anion, thereby increasing the dissociation of the Li salt.

The SiO_2@BNNS nanomaterial was dispersed in anhydrous acetonitrile, as shown in Figure S4a. Then PEO and LiTFSI were mixed into the solution (shown in Fig. S4b). After drying, an independent CPE film with good flexibility and mechanical strength can be obtained, as shown in Figure S5. The images in Figure S5 show a folded CPE film that could quickly restore its original shape without breaking, showing the excellent flexibility robustness necessary for battery design. Figure 1e shows the surface morphology of PEO/LiTFSI/4 wt%SiO_2@BNNS CPE. The sharp bright spots are the nanoparticles on the surface of the electrolyte, while the fuzzy spots are the particles below the surface. The nano fillers are uniformly dispersed in the electrolyte, and the particle size distribution was uniform. The inset is an electronic image of the prepared flexible film. The SEM cross-sectional image in Figure 1f and the image in

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Figure S6 show that the thickness of PEO/LiTFSI/4 wt% SiO_2@BNNS CPE was about 110 μm.

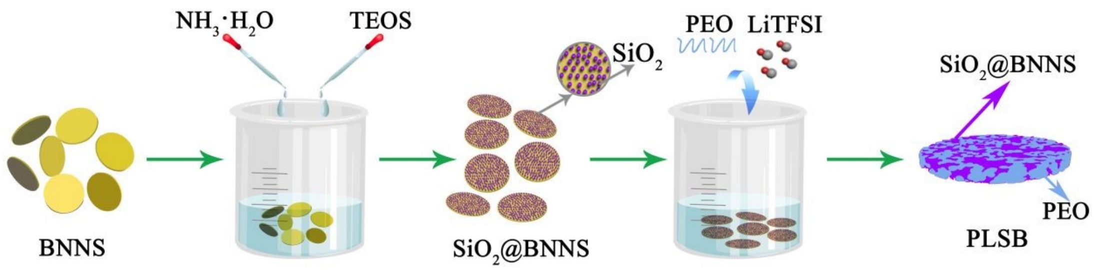

Figure 1. Schematic illustration of the preparation of PEO/LiTFSI/SiO_2@BNNS composite solid electrolyte membrane.

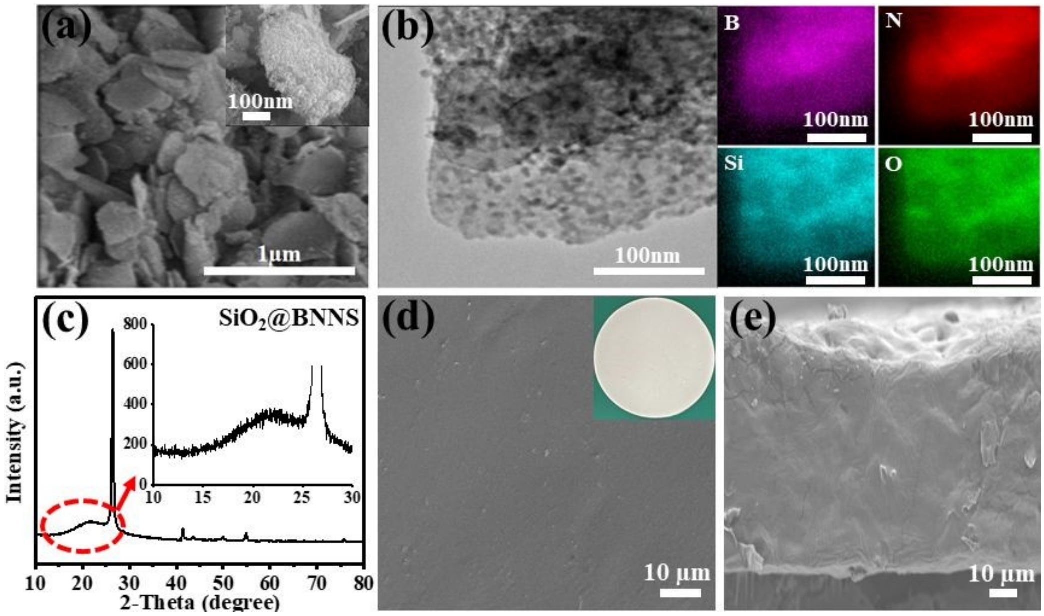

Figure 2. (a) The SEM image of SiO_2@BNNS. The inset figure shows partial enlarged image. (b) The TEM and Elemental mapping images of SiO_2@BNNS. (c) X-ray diffraction data of SiO_2@BNNS composite. The inset figure shows partial enlarged image between 10-30° part. (d) top view image of PEO/LiTFSI/SiO_2@BNNS CPE. The inset is photograph of the flexible thin film. (e) cross-sectional image of PEO/LiTFSI/SiO_2@BNNS.

The XRD data of h-BN, SiO_2@BNNS, pure PEO, LiTFSI, PEO/LiTFSI and PLSB CPE are revealed in Figure 3a and Figures S7 and S8. The h-BN spectrum in Figure S7 shows the characteristic diffraction peaks on the (0 0 2), (1 0 0), and (0 0 4) planes.[41, 42] There are many sharp diffraction peaks in the XRD spectra of LiTFSI in Figure S8, which match the JCPDS card (No. 01-082-2341), indicating its high degree of

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crystallinity at room temperature. The sharp peaks at 19° and 23° in the XRD spectra (green line) shown in Figure 3a can confirm the crystalline characteristics of pure PEO. However, there are no obvious LiTFSI characteristic peaks in the XRD spectra of PEO/LiTFSI and PLSB CPE, which can be attributed to LiTFSI decomposed into Li^+ and $TFSI^-$ ions. Furthermore, due to the addition of lithium salt, the intensity of the PEO peak in the SPE is decreased compared to that in the pure PEO phase, showing that the addition of lithium salt destroyed the crystallinity of PEO. The high degree of delocalization of the anionic electron cloud in the lithium salt can increase the mobility of the polymer chain. The electron withdrawing effect of $-SO_2CF_3$ favoring the negative charge on the N atom being dispersed, shielding the entire molecule and reducing the combination of anions and surrounding cations, playing a plasticizing role and improving conductivity. As the content of SiO_2@BNNS gradually increased, the relative intensity of the PEO peak in PLSB decreased further, as shown in Figure 2b, which confirms that the addition of SiO_2@BNNS can further reduce the crystallinity of PEO. The OH groups on the SiO_2 surface are expected to favor interactions (via hydrogen bonding) with the PEO segments, with a resulting increase in the local PEO amorphous phase fraction, the decreasing crystallinity of PEO dramatically enhances the segmental motion of polymer chains, which strengthen the binding between Li^+ and ether groups on PEO chains, which contributes to the separation of Li^+ and $TFSI^-$ in PEO matrix. The surface interaction between SiO_2@BNNS and PEO/LiTFSI can also prevent the polymer from recrystallization at room temperature.

Ionic conductivity is very important for evaluating Li^+ ion mobility in a CPE.[43, 44] In order to obtain the best ionic conductivity, the CPE containing different SiO_2@BNNS additives was sandwiched between two stainless-steel electrodes for AC impedance spectroscopy measurement. The results are shown in Figure 2c–e. Figure 2c shows the impedance curve of the PEO/LiTFSI/xSiO_2@BNNS (x is the weight ratio of SiO_2@BNNS to PEO) sample at 60 °C. The total resistance of the membrane corresponds to the intercept value of the low-frequency line on the x-axis. To be sure, all PEO/LiTFSI/xSiO_2@BNNS membranes have higher ion conductivity than PEO/LiTFSI membranes. Due to the decrease in crystallinity, the proportion of

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amorphous area increased, so more conducive to the transport of Li^+. The EIS plot of Figure S9 further proves the effect of SiO_2 nanoparticles uniformly distributed on the surface of boron nitride. SiO_2 nanoparticles are distributed evenly on the surface of the boron nitride, which ingeniously not only alleviates the problem of nanofiller agglomeration but also increases the contact area with the electrolyte matrix. Therefore, it can promote the interaction with the Lewis acid-base of the anion, thereby benefit for the dissociation of the lithium salt to improve Li−ion conductivity.[23] The non-monotonic behavior of ionic conductivity is understandable, because a high content of nanofillers will cause aggregation[45] and free volume depletion,[32] which will further reduce ion percolation and hinder ion transmission. [46, 47] Subsequently, the influence of temperature on ionic conductivity was explored. The sample containing with 4 wt% SiO_2@BNNS nanofiller had an ionic conductivity of 4.53×10^{-4} S cm^{-1} at 60 ℃, in line with those of other reported ion-conducting oxides/sulfides, such as garnet $Li_7La_3Zr_2O_{12}$,[48, 49] perovskite oxides.[49] Figure 3e shows the effect of temperature on the ion conductivity. The temperature of the PEO/LiTFSI/4wt%SiO_2@BNNS film was increased from 30 °C to 80 °C, and the impedance decreased accordingly. All curves can be fitted using a typical Vogel–Tam man–Fulcher (VTF) model[5], which confirmed the amorphous nature of the PEO/LiTFSI/xSiO_2@BNNS membrane. PEO will melt above 60 °C, while the polymer electrolyte with SiO_2@BNNS still has higher ionic conductivity, which indicates that the interaction between SiO_2@BNNS and the polymer electrolyte not only increases the ionic conductivity but also improves the thermal stability of the electrolyte. This will be discussed in a later section. The structural stability of CPE is also an important factor in the evaluating polymer electrolytes, because good mechanical strength can achieve, to a certain extent, the purpose of inhibiting the formation and growth of lithium dendrites. Therefore, we conducted a mechanical performance measurement based on the strain-stress curve. From Fig. 3f, the maximum values of stress and strain of CPE with SiO_2@BNNS were 1.33 MPa and 400%, respectively. This shows that the introduction of SiO_2@BNNS does improve the mechanical properties of CPE. In contrast, due to the low Young's modulus,[50] the maximum stress of CPE membrane without SiO_2@BNNS was only

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0.87 MPa. The enhanced mechanical strength is due to the SiO_2@BNNS incorporated into the PEO polymer matrix.[51]

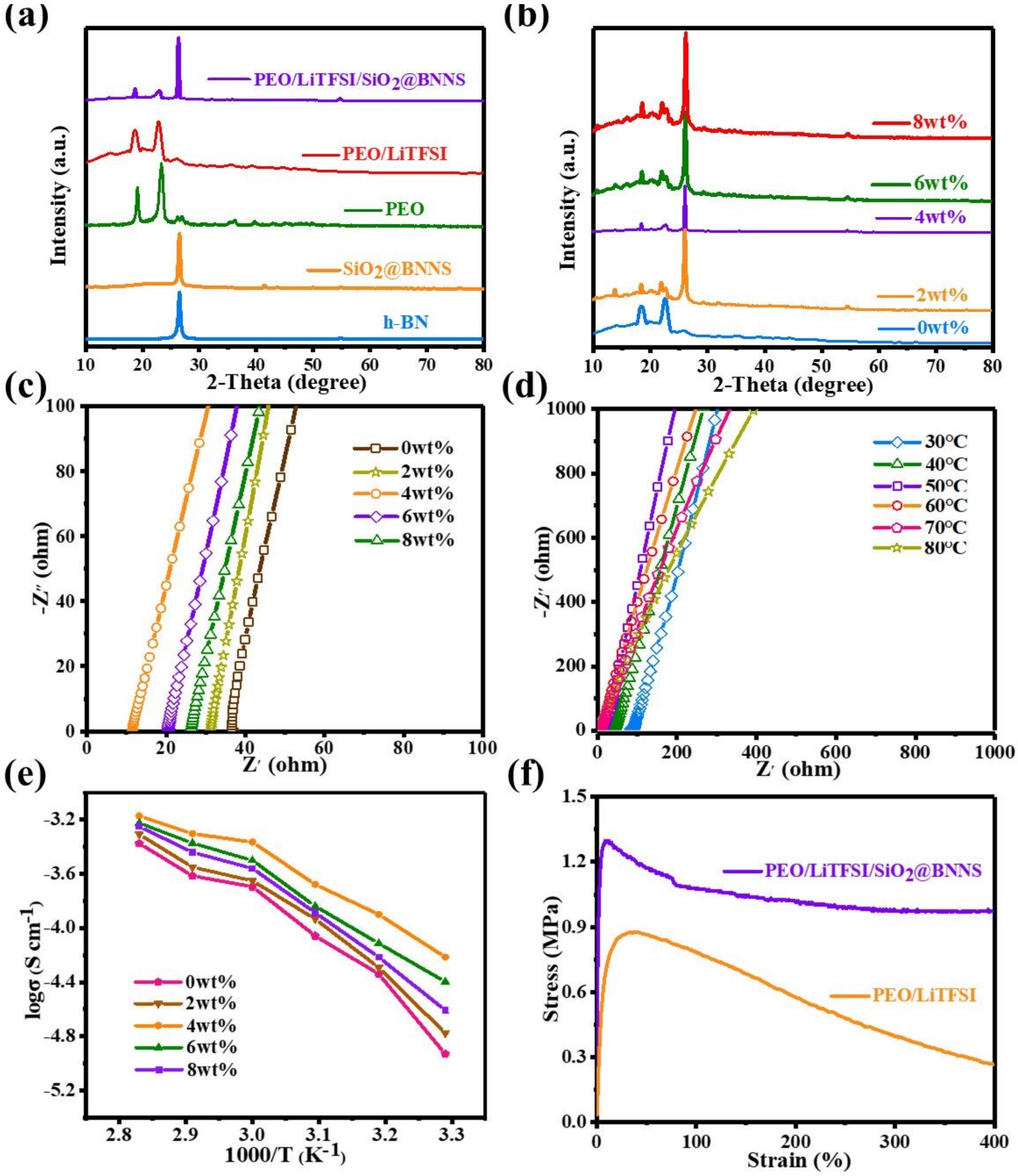

Figure 3. (a) X-ray diffraction data of h-BN, SiO_2@BNNS, PEO, PEO/LiTFSI SPE and PEO/LiTFSI/4wt% SiO_2@BNNS CPE. (b) X-ray diffraction results of CPE-xSiO_2@BNNS membranes (x= 0%, 2%, 4%, 6% and 8%). (c) The EIS plots of CPE-xSiO_2@BNNS membranes at 60℃ (x= 0%, 2%, 4%, 6% and 8%), where Z′is the real part of impedance, Z′′is the imaginary part of impedance. (d) The EIS plots of impedance at different temperature. (e) Ionic conductivity as a function of temperature for CPE-xSiO_2@BNNS membranes (x= 0%, 2%, 4%, 6% and 8%). (f) The stress-strain curves of PEO/LiTFSI/4wt% SiO_2@BNNS CPE and PEO/LiTFSI SPE.

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In order to verify that the addition of SiO_2@BNNS improves the thermal stability of the polymer, TGA was carried out. In Figure 3a, we can see the thermogravimetric curve of PEO/LiTFSI electrolyte and PLSB CPE at 25−800 °C. Below 100 °C, a weight loss of 1.6 wt% can be observed, which is due to the loss of absorbed water. The maximum weight loss of PEO/LiTFSI occurs at 360−440 °C, which corresponds to the decompose of PEO and LiTFSI. The remaining amount of PEO/LiTFSI electrolyte was about 4% by weight. The PLSB CPE showed higher decomposition temperatures at around 450 °C compared to PEO/LiTFSI electrolyte (366 °C), which shows that adding SiO_2@BNNS to SSE improves the thermal stability of the CPE. This is because the strong shielding effect of the BNNS significantly delays the escape of degradation products of PEO and LiTFSI, and the surface modification of h-BN enhances the interface interaction between adjacent h-BNs, improving the thermal conductivity and thereby greatly enhancing the thermal stability. Furthermore, Figure 4b shows the thermal stability of Celgard 2300, PEO/LiTFSI and PLSB CPE at 30, 50, 100 and 150 °C. When the temperature was increased to 50 °C, Celgard 2300 and PEO/LiTFSI SPE showed slight signs of heat contraction, but the shape of the PLSB CPE was well preserved. When the temperature increased further to 100 °C, the original state of the PLSB CPE was still retained. In contrast, melting behavior was observed in PEO/LiTFSI, and Celgard 2300 contracted further due to softening of the polypropylene. When the test temperature was increased to 150 °C, Celgard 2300 contracted quickly, and PEO/LiTFSI reached an almost completely molten state; this could cause the battery to short-circuit or even explode at high temperatures. From 30 °C to 150 °C, the PLSB CPE maintained its original form, which shows definitively that the thermal stability of the CPE was greatly improved.

To further evaluate the heat dissipation ability and difference in thermal conduction between the different samples, the thermography and temperature changes of PEO/LiTFSI, PEO/LiTFSI/h-BN and PEO/LiTFSI/SiO_2@BNNS were recorded every 5s by an IR camera (Fig. 3c). PEO/LiTFSI exhibited the slowest temperature change and highest temperature, while PEO/LiTFSI/h-BN showed a faster, and PEO/LiTFSI/SiO_2@BNNS the fastest temperature change. The color distribution of

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PEO/LiTFSI/SiO_2@BNNS was more even than PEO/LiTFSI/h-BN composites, despite the similar temperature curve. Compared with h-BN, the reduced interface thermal resistance of SiO_2@BNNS helps to increase the thermal conductivity of SiO_2@BNNS, and the surface modification of h-BN enhances the interface interaction between adjacent h-BNs improves the thermal conductivity. Therefore, the increase in thermal conductivity leads to enhanced thermal stability, which provides support for battery operation in high-temperature environments.

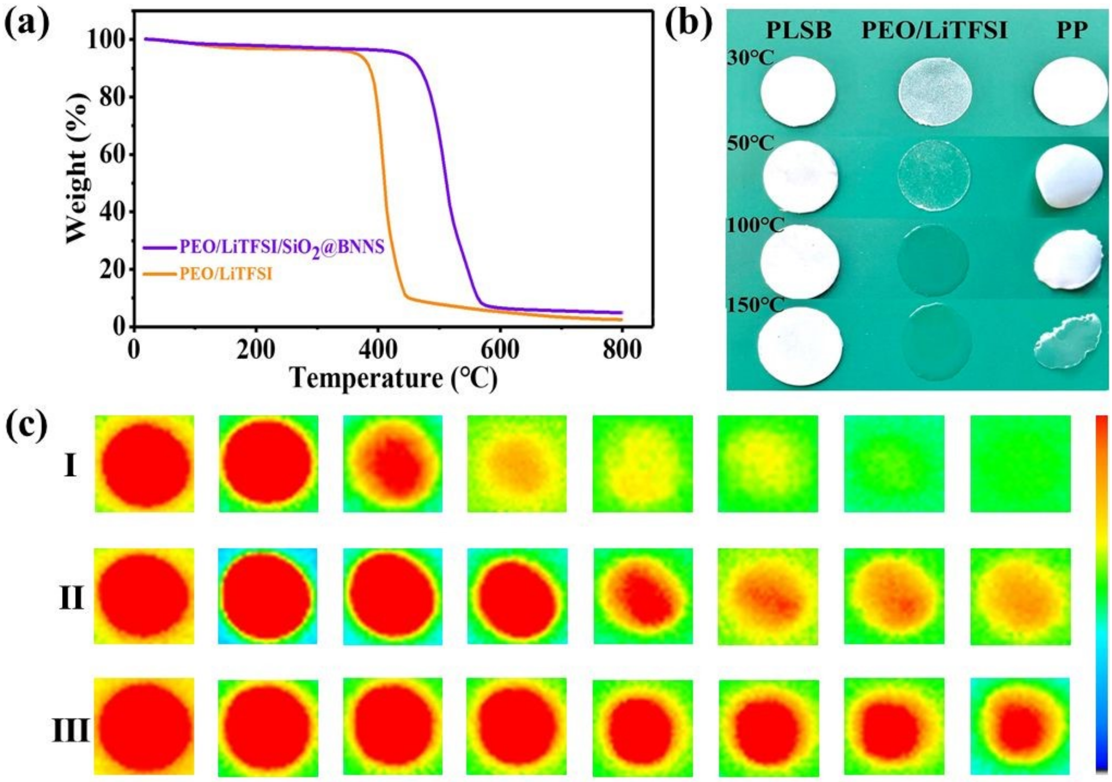

Figure 4. (a) TGA curves of PEO/LiTFSI and PEO/LiTFSI/4wt%SiO_2@BNNS. (b) Photographs of PEO/LiTFSI/4wt%SiO_2@BNNS, PEO SPE and PP separate after heat treatment at various temperatures for 15 min. (c) infrared thermal images of different samples (I PEO/LiTFSI/SiO_2@BNNS; II PEO/LiTFSI/h-BN; III PEO/LiTFSI).

By adding SiO_2@BNNS, the ionic conductivity of CPE containing SiO_2@BNNS is effectively improved, which can be explained by the following two aspects: First, the well distributed SiO_2 has a large effective surface area, which may promote the interaction with the Lewis acid-base of the anion. The OH groups on the SiO_2 surface are expected to favor interactions (via hydrogen bonding) with the PEO segments, with a resulting increase in the local PEO amorphous phase fraction, the decreasing

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crystallinity of PEO dramatically enhances the segmental motion of polymer chains, which strengthen the binding between Li^+ and ether groups on PEO chains. Both of them contribute to the separation of Li^+ and $TFSI^-$ in PEO matrix (Schematic 1). This limits the transport of anions, increasing the Li^+ ion migration number of CPE, and also inhibiting the growth of dendritic Li. To confirm this hypothesis, FT-IR tests were performed. Figure S10 and S11 show the FT-IR spectra of SiO_2@BNNS between 4000 and 500 cm^{-1} and also the spectra of PEO/LiTFSI and PLSB within 1400–1180 cm^{-1}. In Figure S10, the characteristic peak of Si-OH and -OH can be observed, the strong Lewis acid-base interaction between electrolyte ion species and the surface chemical groups of ceramic fillers can enhance salt dissociation and stabilize the anions. [52] In Figure S11, peaks at 1186, 1205, and 1230 cm^{-1} can be observed in both PEO/LiTFSI and PLSB electrolytes, which correspond to the asymmetric and symmetric stretching of -CF_3. Other peaks of PEO can be observed at 1342 and 1280 cm^{-1}. In addition, peaks corresponding to asymmetric -SO_2- stretching are observed at 1327 and 1298 cm^{-1} in the PEO/LiTFSI electrolyte. However, in the PLSB electrolyte the peaks shifted to 1294 and 1333 cm^{-1}. The movement of specific peaks in FT-IR is caused by the interaction between SiO_2@BNNS and –SO_2– in the $TFSI^-$ anion, which indicates that the dissociation of the lithium salt is enhanced, and more Li^+ is released, which further improves ionic conductivity. The added SiO_2@BNNS can reduce the crystallinity of PEO and inhibit the recrystallization of the PEO chain, which will enhance the segmented movement of the polymer and further promote the migration of lithium ions.

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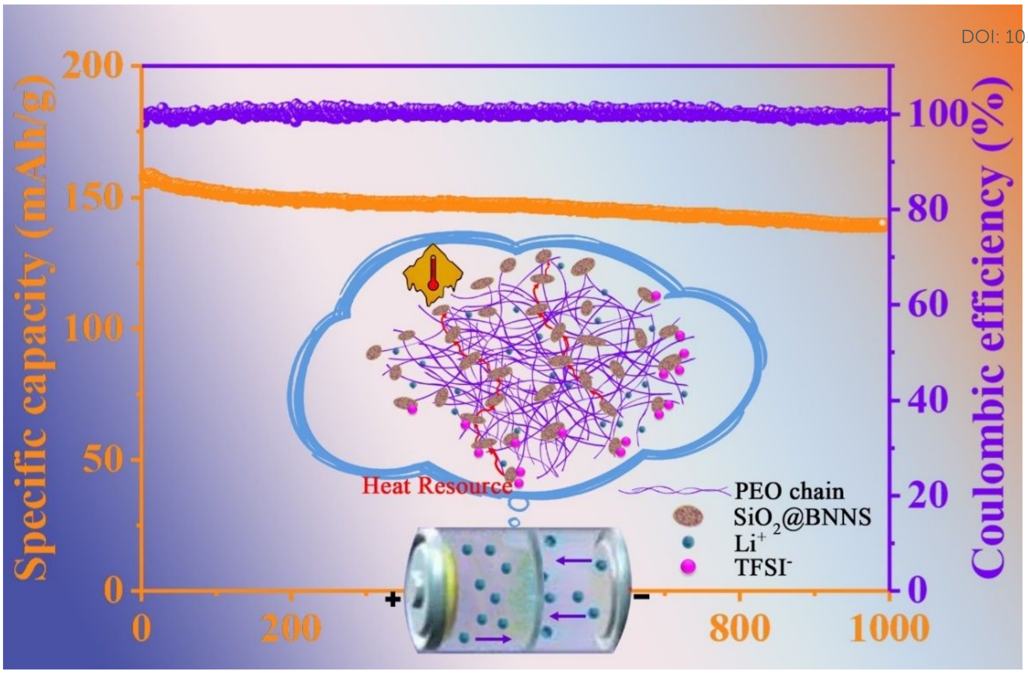

Schematic 1. Schematic illustration of mechanism for enhanced Li ion transport and heat conduction in PEO-based electrolyte by SiO_2@BNNS.

In addition to thermal and chemical stability, the PLSB CPE can also maintain stability with Li metal anodes and has a positive effect on alleviating the formation and growth of Li dendrites.[53-56] As seen in Figure 5a, galvanostatic cycling measurements were made with a current density of 0.4 mA cm^{-2}. It can be seen that the Li plating and stripping overpotential of the PLSB CPE of the battery was close to 120 mV at 0.4 mA cm^{-2} and remained steady for 1000 h. No short circuits or other detectable failures occurred, which can be attributed to the increase in mechanical strength and Li-ion transference. The inherent mechanical strength of SiO_2@BNNS improves the electrolyte's ability to inhibit lithium dendrites, and the modification of the surface of BN by SiO_2 nanoparticles increases the specific surface area, increases the number of interaction sites between Lewis acid and anion base improves the Li-ion transfer rate and decreases the polarization voltage. In contrast, the battery with PEO/LiTFSI SPE showed a high overpotential of 1000 mV at the beginning of the cycle and the voltage curve is extremely unstable, which indicates the instability of PEO/LiTFSI SPE. All Li/Li symmetry battery tests confirmed that the prepared CPE using SiO_2@BNNS has excellent stability to lithium metal anodes. To explore the stability of CPE under high

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pressure, we measured the electrochemical window of the sample at 60 °C using a stainless-steel/CPE (PLSB and PEO/LiTFSI)/Li battery (Figure 5d). At a scan rate of 1 mV s^{-1}, the voltage of the electrochemical window of the CPE containing 4 wt% SiO_2@BNNS was around 4.71 V, which is better than the SSE (4.25 V) without SiO_2@BNNS. The enhancement of the electrochemical stability of the CPE indicates that these CPEs may be well matched with high-voltage cathode materials and do not decompose under high voltage. Therefore, the electrolyte can match the Li metal anode and the high-voltage cathode to form a rechargeable lithium battery with higher energy density.

We also tested and calculated the Li-ion migration number in two electrolytes (PEO/LiTFSI SPE and PLSB CPE) by chronoamperometry and AC impedance measurement. Figure 5c shows the IT curve and AC impedance spectra of the Li/PLSB CPE/Li battery before and after DC polarization at 60 °C. There was almost no change in the interface resistance of the Li/PLSB CPE/Li battery before and after DC polarization. According to the Bruce–Vincent equation, it could be calculated that the t_{Li+} of PEO/LiTFSI SPE is only 0.19. In contrast, PLSB CPE exhibited a high t_{Li+} of 0.54. The amelioration of t_{Li+} may benefit from the interaction of $TFSI^-$ anion and SiO_2@BNNS. The dissociation of $TFSI^-$ anion by SiO_2@BNNS promotes local relaxation and segmental movement, thereby generating more free Li^+ ions,[13] increasing the number of Li-ion migration, which was conducive to the transmission of Li^+ ions.

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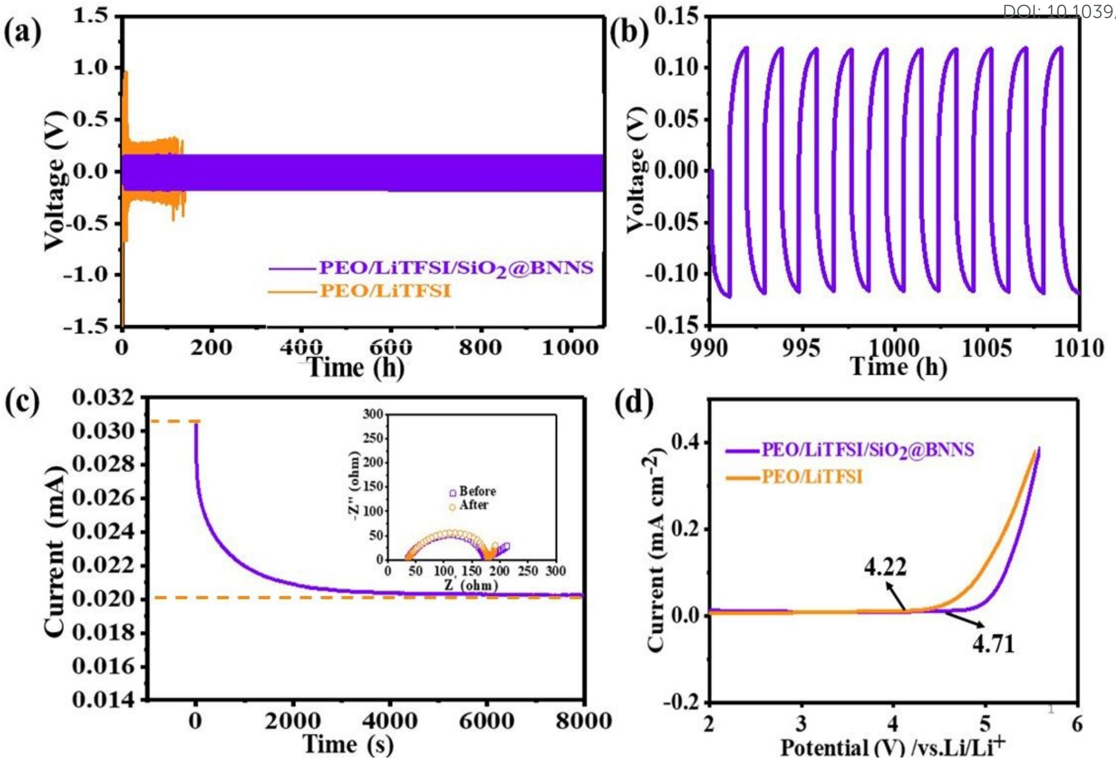

Figure 5. (a) Galvanostatic cycling of Li symmetric battery using PEO/LiTFSI/4wt% SiO_2@BNNS CPE and PEO/LiTFSI SPE at the current density of 0.4 mA cm^{-2}. (b) The tail region of Galvanostatic cycling of Li symmetric battery using PEO/LiTFSI/4wt% SiO_2@BNNS CPE for selected cycles. (c) Chronoamperometry curve for PEO/LiTFSI/4wt% SiO_2@BNNS CPE at 60 °C with polarization voltage of 10 mV and EIS plots before and after the polarization inset. (d) Linear sweep voltammetry of (LSV) of PEO/LiTFSI SPE and PEO/LiTFSI/4wt% SiO_2@BNNS CPE at 60 °C under the scan rate of 1 mV.

In order further to confirm the excellent ability of PLSB to alleviate lithium dendrite growth, the surface morphology of the lithium anode obtained after different cycle times was analyzed. Figures 6a, b and c respectively show the morphological changes of the lithium metal anode in the Li/PLSB/Li symmetric battery after cycle times of about 50 h, 150 and 750 h at a current density of 0.4 mA cm^{-2}, in which a dense, flat, and smooth surface can be observed. In contrast, the Li anode of Li/PEO/Li symmetric battery cycled about 50 h (Figure 6d), 100h (Figure 6e), 150h (Figure 6f) with current densities of 0.4mA cm^{-2} show a large number of irregular island-shaped lithium dendrites. And as the cycle time increases, the growth of lithium dendrites becomes more and more intense, which would eventually cause a short−circuit in the

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battery. The excellent ability of PLSB to inhibit the growth of lithium dendrites may be ascribed to two reasons. First, as shown in Figure 1e, SiO_2@BNNS is distributed evenly inside the solid electrolyte. Due to the high strength of SiO_2@BNNS, the overall mechanical strength of PLSB is improved. As a result, the growth of lithium dendrites will be effectively suppressed. Secondly, SiO_2@BNNS is embedded uniformly in the polymer matrix, which reduces the crystallinity of PEO and improves the Li^+ ion conductivity, thereby reducing the growth of lithium dendrites.

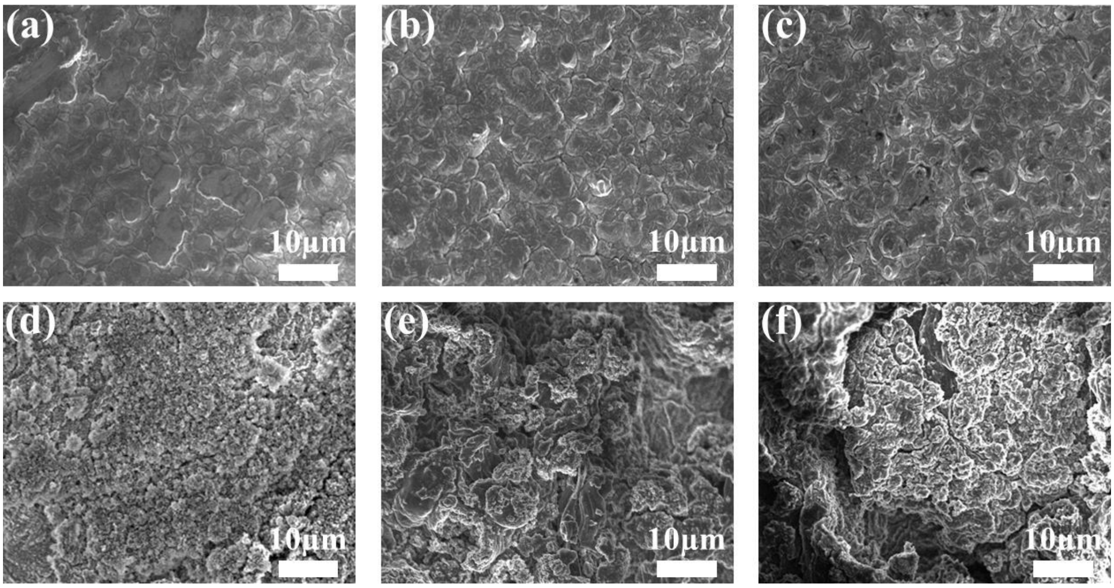

Figure 6. The surface morphologies of Li electrode obtained from a) Li/PLSB/Li symmetric cells cycling about 50 h; b) the surface morphology of Li electrode obtained from Li/PLSB/Li symmetric cell cycling about 150 h; c) the surface morphology of Li electrode obtained from Li/PLSB/Li symmetric cell cycling about 750 h; the surface morphologies of Li electrode obtained from d) Li/PEO/Li symmetric cells cycling about 50 h; e) the surface morphology of Li electrode obtained from Li/PEO/Li symmetric cell cycling about 100 h; the surface morphology of Li electrode obtained from Li/PLEO/Li symmetric cell cycling about 150 h at 60 °C.

In order to study further the compatibility between the PLSB film and lithium metal, a lithium symmetrical battery was cycled at various current densities and temperatures (Figure 7). When cycled at 25 °C (purple line in Figure 7a) with a stripping/plating capacity of 0.02 mAh/cm², the voltage distribution of the battery showed stability for more than 180 h. However, PEO/LiTFSI SPE (yellow line in Figure

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7a) showed an extremely unstable voltage curve at 25 °C, the polarization voltage also increased greatly, and the battery was short-circuited at about 100 h. At 60 °C (Figure 7b), the observed voltage increased slowly with increasing current density, corresponding to the establishment of a salt concentration gradient on the cells. When the current density was increased to 0.4 mA cm^{-2}, the voltage increased to 110 mV. As the current density decreased successively, the overpotential returned to the initial value, which proves the good cycle stability of PLSB CPE. For the battery using PEO SPE, when the current density increased to 0.1mA cm^{-2}, the voltage increased suddenly to more than 0.1 V (Figure 7c), which indicates that the concentration of Li^{+} on the anode surface is close to zero and the limiting current density is reached. As the temperature increase will cause the ion conductivity to increase, the overpotential obtained from cycling the lithium symmetric battery also decreases with the increase in temperature. When the battery was cycled at 150 °C and 0.4 mA cm^{-2} (Figure 7d), the battery lasted for more than 800 h, and the voltage distribution of the battery was also greatly reduced compared to the voltage at 60 °C. PLSB CPE greatly improves the thermal stability of LMBs. These results show that even at different temperatures, the PLSB CPE film has excellent compatibility with lithium metal.

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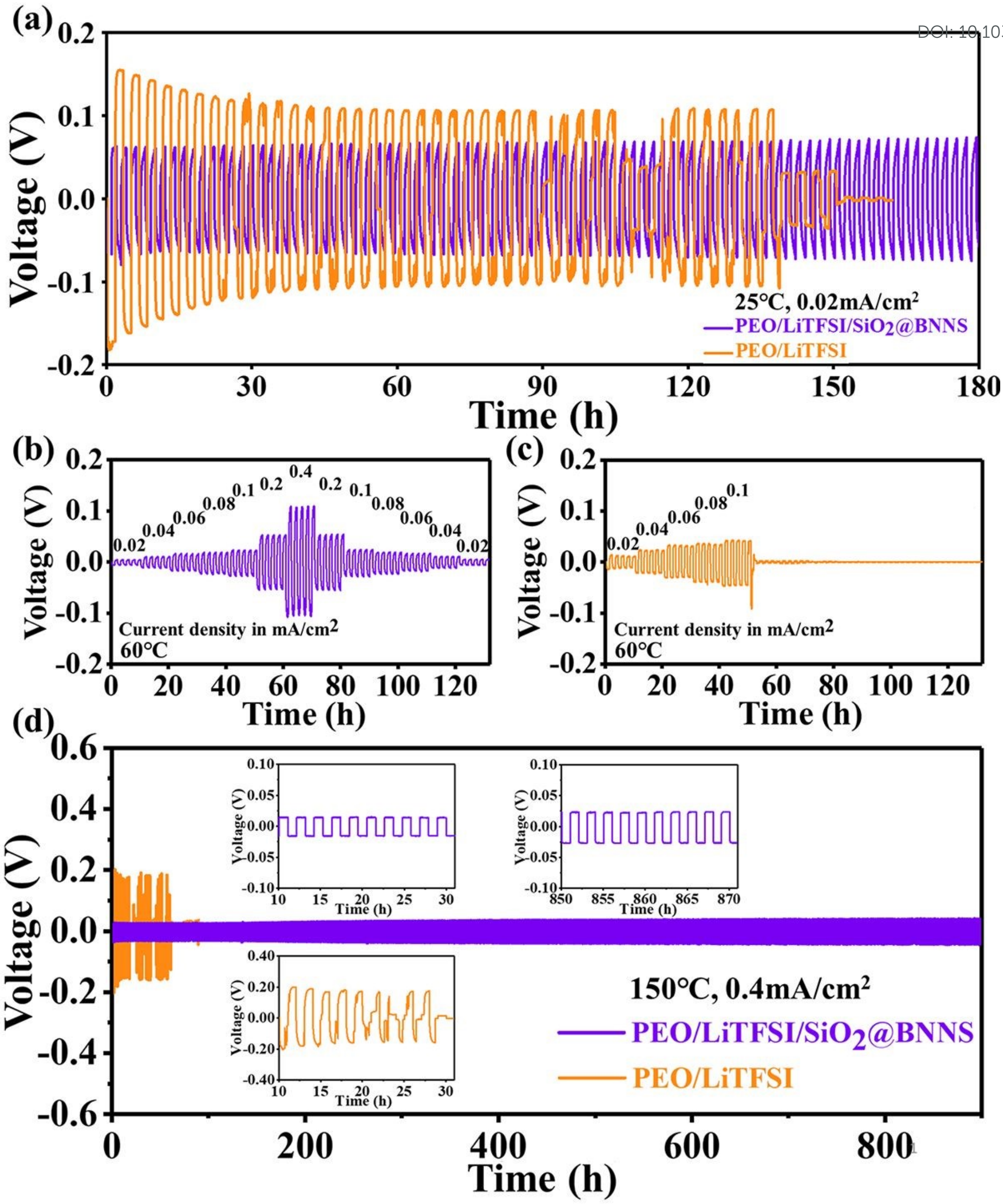

Figure 7. Lithium symmetric cell performance. Voltage profiles of symmetric cells cycled at a) 25 °C, b) and c) 60 °C with increasing steps of current density. The charge and discharge time is 1 h, respectively, and the current density (labeled above each step in mA/cm^2) is stepped every 5 cycles. (d) Long term voltage profiles of symmetric cells cycled at 150 °C using a current density of 0.4 mA/cm^2.

In order to verify the performance of PLSB in ASSLMBs, an all-solid-state battery with PLSB CPE as electrolyte, lithium metal as anode and $LiFePO_4$ as cathode was prepared. The electrochemical cycling performance of the $LiFePO_4$/Li full battery using PEO/LiTFSI/4 wt% SiO_2@BNNS CPE as the electrolyte is shown in Figure 8a.

At a current density of 0.2 C, the battery using PEO/LiTFSI/4 wt% SiO_2@BNNS displayed higher initial coulombic efficiency (94%) because of the higher mobility of Li^+ ion in PEO/LiTFSI/4 wt% SiO_2@BNNS. After 150 cycles, it still showed a high capacity retention rate (93%). However, the initial coulombic efficiency of the battery using PEO/LiTFSI as the electrolyte was 92%. It showed a very poor capacity retention rate, with a specific discharge capacity of only 22.7 mAh g^{-1} after 150 cycles, which is due to the poor mechanical strength and low Li^+ ion mobility caused by the high resistance. The performance when paired with a high-load cathode (> 1.0 mAh cm^{-2}) has also been studied, as shown in Fig S12 and Fig S13, when matched with 1 mAh/cm^2 LEP cathode, the specific capacity was 75.8 mAh/g after 50 cycles at a current density of 0.2 C. When matched with 1.3 mAh/cm^2 LEP cathode, the specific capacity was 50.9 mAh/g after 50 cycles at a current density of 0.2 C. Figure 8b shows the charge and discharge behavior of SSLMB with PLSB CPE at different current densities. At 0.2 C, the charging and discharging voltage platforms were approximately 3.50 V and 3.35 V, respectively. As the current density gradually increased, the overpotential between the charging and discharging platforms increased slightly. This indicates that PLSB can provide a stable interface for Li^+ ion transport during battery cycles. In addition, the charge and discharge behavior of SSLMB with PLSB CPE at different temperatures was also measured (Fig S14). As the temperature increased gradually, the transmission speed of lithium ions increased, the overpotential between the charging and discharging platforms decreased slightly, which indicating the fast kinetics of Li^+ transport in SPE. Figure 8c shows the rate performance of SSLMBs using different electrolytes (PLSB CPE and PEO/LiTFSI SPE). The specific discharge capacities of SSLMB (PLSB CPE) at 0.1, 0.2, 0.5 and 1.0 C were 158, 140, 124 and 117 mAh g^{-1}, respectively. When the current density returned to 0.2 C, the discharge capacity could still reach about 160 mAh g^{-1}, indicating that PLSB CPE has excellent cycle performance. In contrast, the specific discharge capacities of the SSLMB (PEO/LiTFSI SPE) at 0.1, 0.2, 0.5 and 1.0 C were 139, 121, 92 and 75 mAh g^{-1}, respectively. When the current density returned to 0.2 C, the specific discharge capacity decreased to 128 mAh g^{-1}. Obviously, whether considering capacity or cycle stability, an SSLMB using a PLSB CPE is better than batteries using a pure PEO/LiTFSI electrolyte. The interface impedance of SSLMB with PEO/LiTFSI/4 wt% SiO_2@BNNS electrolyte and PEO/LiTFSI electrolyte was also measured by EIS. As shown in Figure 8d, the semicircle in the high frequency region represents the charge transfer resistance (R_{ct}), and the intersection of the real axis at the high frequency represents the ohmic resistance (R_o).[57] In Figure 8d, the R_{ct} of PEO/LiTFSI/4 wt% SiO_2@BNNS is 478.3 and R_o is 317.1 Ω, which is smaller than the R_{ct} (550.1 Ω) and R_o (359.1 Ω) of PEO/LiTFSI. This result further verified the higher ionic conductivity of PEO/LiTFSI/4 wt% SiO_2@BNNS, and again confirmed the positive effect of SiO_2@BNNS.

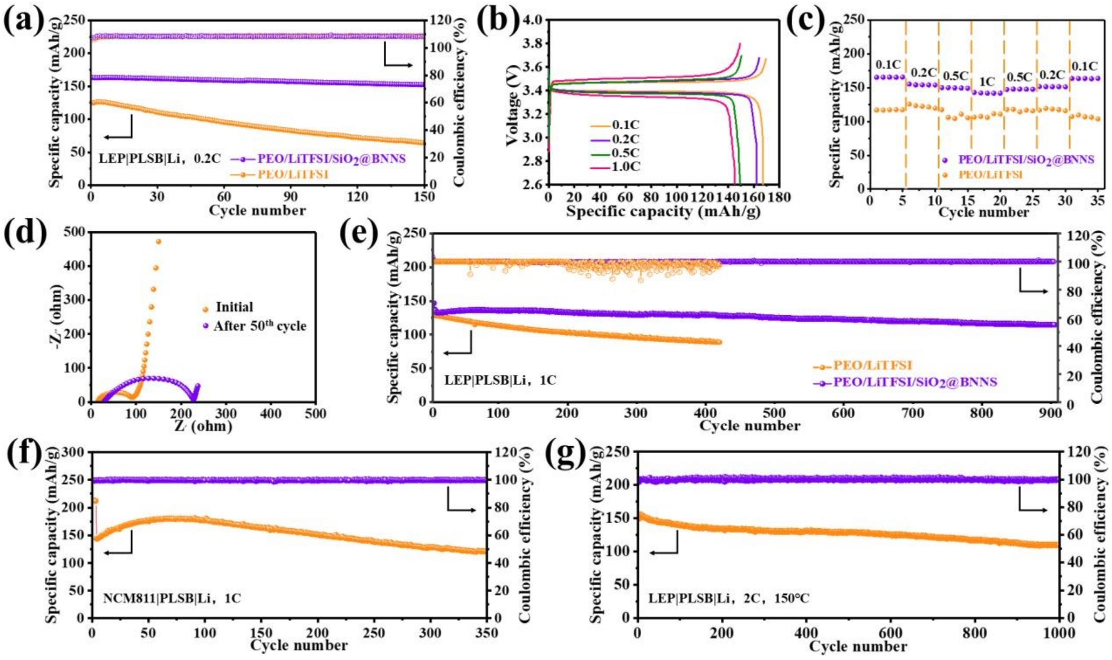

Figure 8. (a) Cycling performance of the solid-state $LiFePO_4$/Li battery at 60 ℃ and 0.2 C. (b) Typical charge−discharge curves of solid-state $LiFePO_4$/Li battery using PEO/LiTFSI/4 wt% SiO_2@BNNS electrolyte at 60 °C. (c) Rate performance of solid-state $LiFePO_4$/Li battery using PEO/LiTFSI/ 4wt% SiO_2@BNNS electrolyte and PEO/LiTFSI electrolyte at 60 °C. (d) Impedance spectra of the solid-state $LiFePO_4$/Li battery using PEO/LiTFSI/4 wt% SiO_2@BNNS electrolyte before and after 50 cycles. (e) Long cycling performance of the solid-state $LiFePO_4$/Li battery at 60 °Cand 1 C. (f) Cycling performance of the solid-state NMC811/Li battery at 60 °Cand 1 C. (g) Long cycling performance of the solid-state $LiFePO_4$/Li battery at 150 °Cand 2 C.

The cycle performance of the $LiFePO_4$/PLSB/Li battery at 1 C and 60 °C is shown in Figure 8e. After 900 cycles, the discharge specific capacity remains at 131mAh/g,

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which is very stable. In contrast, the $LiFePO_4$/Li battery using PEO/LiTFSI/BNNS as the electrolyte has a higher interface impedance, a lower initial specific capacity, and the content retention rate drops below 80% after 150 cycles.(Fig S15) The cycle stability and thermal stability of the electrolyte are greatly reduced, which proves the positive effect of SiO_2 nanoparticles. The LSV shown in Figure 5d shows that PLSB has increased the electrochemical window to 4.71 V, so matching the NMC811 cathode to test its stability to the high-voltage cathode materials, as shown in Figure 8f, it shows stable cycle performance within 350 cycles. The initial specific capacity reaches 222.8 mAh/g. The cycle performance at 2 C and 150 °C showed good stability at high temperature, as shown in Figure 8g, the capacity retention rate was 87% after 1000 cycles. When the current density was increased to 3 C, the battery can still be cycled stably 500 times, and the coulombic efficiency was still close to 100% (Fig S16). This remarkable electrochemical performance is attributed to the outstanding thermal and electrochemical stability of PLSB CPE.

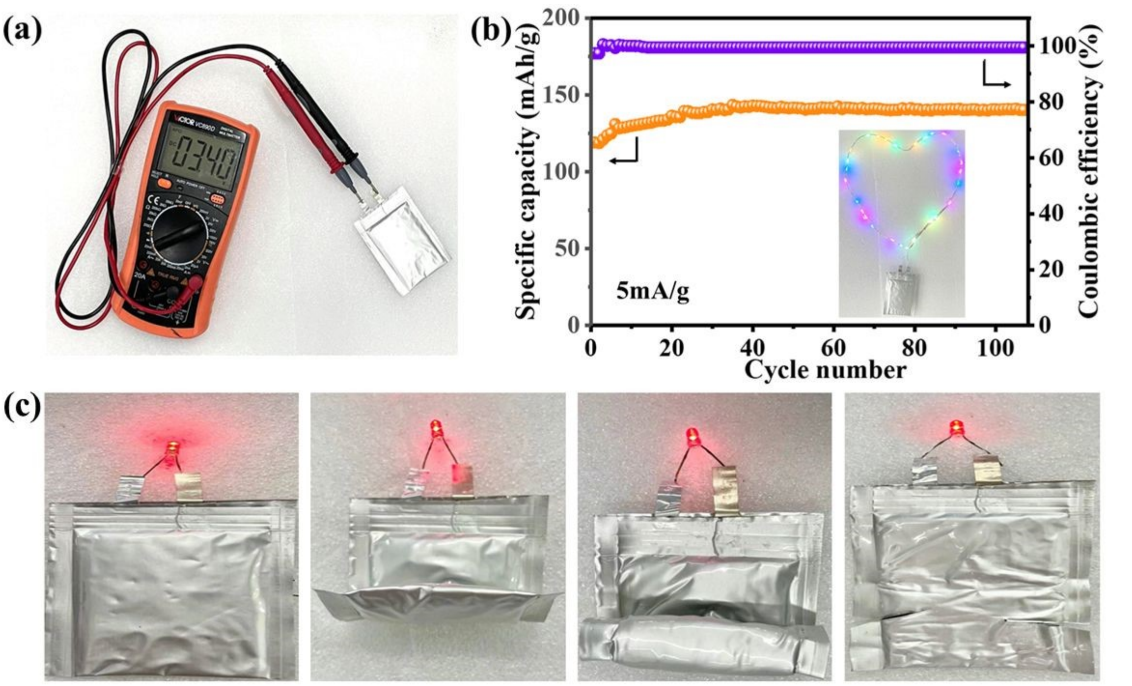

Figure 9. (a) The digital photograph of $LiFePO_4$/Li pouch battery with PEO/LiTFSI/4wt% SiO_2@BNNS electrolyte. (b) Cycling performance of Li/$LiFePO_4$ SSLMBs with PEO/LiTFSI/4wt% electrolyte at the current density of $5mA/cm^2$ and 60 °C. The inset is a digital photograph of LEDs

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lightened by SSLMBs. (c) Flexibility and safety evaluation of $LiFePO_4$/Li pouch batteries.

The SSLMBs showed a voltage of 3.40 V, as shown in Figure 9a. Figure 9b shows that when the current density of SSLMB was 5 mA/cm^2 at 60 °C, it exhibited a stable specific capacity close to 140 mAh g^{-1} during 100 discharge and charge cycles, the coulombic efficiency was close to 100%. These excellent electrochemical properties indicate that the PLSB CPE exhibits good interface stability and ion conductivity in all solid-state batteries. The inset in Figure 9b confirms that such an SSLMB can light up a "heart−shaped" LED, showing its potential for practicality. As shown in the Figure 9c, after being folded and cut, the pouch battery can still power the LED light, indicating that it can still work well under various mechanical abuses, attributable to the good flexibility and stability of PLSB CPE.

CONCLUSION

A PEO-based solid composite electrolyte with SiO_2@BNNS as filler was prepared by a simple and convenient method. Compared to PEO solid electrolyte, the prepared PLSB CPE showed higher thermal stability and higher ionic conductivity. The surface modification of BNNS enhances the interface interaction between BNs, thereby reducing the thermal resistance, and can effectively improve the thermal conductivity with a relatively small filler content, which greatly improves the thermal stability of the CPE. In the meantime, the surface modification of h-BN expands the effective specific surface area, can promote the interaction with the Lewis acid-base of the anion, thereby breaking the interaction between Li^+ and the Li salt anion and improving the Li^+ mobility. In addition, due to the mechanical strength of SiO_2@BNNS, the growth of lithium dendrites is also effectively alleviated. Therefore, the SSLMB assembled with PLSB CPE could be cycled stably for more than 900 cycles with an initial specific capacity of 150 mAh g^{-1} at 1 C. The Li/Li symmetrical battery with PLSB CPE showed a long cycle performance of 900 h at 0.4 mA cm^{-2}. At a high temperature of 150 °C, the battery could cycle stably for more than 1000 cycles at a rate of 2 C. In addition, the safety test of the pouch battery further illustrated the high stability and safety of a CPE based on SiO_2@BNNS.

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